FIT AR Navigation App (FITARNA)

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Motivation

- Navigation Complexity: Large-scale academic facilities like the Evans Library present significant wayfinding challenges for new students and visitors.
- GPS Limitations: Standard satellite-based navigation fails indoors due to signal attenuation and lack of floor-level granularity.
- The Solution: FITARNA leverages Augmented Reality to provide intuitive, real-time visual guidance, bridging the gap between digital maps and the physical environment.

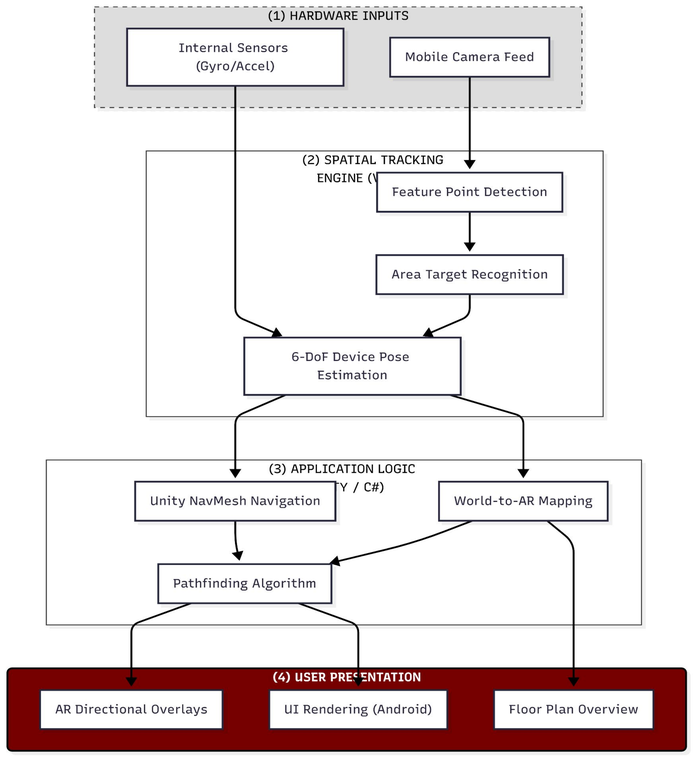
Project Goals

- High-Precision Localization: Achieve sub-meter indoor positioning accuracy without the use of GPS by leveraging AR spatial mapping.
- Intuitive AR Wayfinding: Develop a seamless navigation experience using real-time 3D directional overlays.
- Dynamic Point-of-Interest System: Implement a searchable database of campus locations, such as specific library wings, study rooms, and help desks.
- Cross-Platform Compatibility: Ensure the application maintains high frame rates and stable tracking across modern mobile AR-capable devices.

Implementation and Approach

- Environment Mapping: Conducted extensive spatial scans of the Evans Library using Vuforia Area Targets to create a high-fidelity digital twin for persistent tracking.
- Hybrid AR Framework: Integrated Unity's AR Foundation with the Vuforia Engine to leverage cross-platform ARCore/ARKit features while maintaining superior object and area recognition.
- Pathfinding Logic: Implemented a custom NavMesh within Unity to calculate real-time, shortest-path routes between navigational waypoints.
- Spatial UI Overlay: Developed a user interface that projects 3D directional markers and "breadcrumbs" onto the physical world, ensuring intuitive wayfinding in a complex indoor environment.

System Architecture



Software Stack

unity

vuforia™

Android

Results

- Centimeter-Level Accuracy: Achieved high-precision localization within X0–X0cm of physical targets in the Evans Library.
- Real-Time Performance: Maintained a consistent XX FPS on mid-range Android devices, ensuring a smooth AR experience.
- Reliable Re-Localization: The system successfully "finds itself" within X seconds of opening the app, even in varying lighting conditions.
- Successful Wayfinding: Completed field tests across the 1st and 2nd floors. *More info here*